

Ehsaan Ali

(AI Engineer)

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Summary

AI/ML Engineer and PhD candidate in Artificial Intelligence & Electrical Engineering at Université de Bretagne Occidentale (Oct. 2025 – Oct. 2028), with a Master's in AI (Gold Medalist, FAST NUCES) and certifications as PIAIC AI Developer and Microsoft Python Expert. Experienced in computer vision, deepfake detection, and image analysis, with expertise in Python, TensorFlow, PyTorch, BERT, RAG, multi-modal architectures, YOLOv8, and Django for deploying AI applications. Currently working on the TWIST Project – AI-based Digital Twin for diagnosing and predicting tidal turbine failures and biofouling, focusing on ML, Deep Learning, Signal Processing, and Computer Vision. Skilled in data preprocessing, model optimization, and delivering scalable AI solutions from research to production.

Education

Doctor of Philosophy (Artificial Intelligence and Electrical Engineering) Université de Bretagne Occidentale, Brest France	Oct. 2025 to Oct. 2028
• Description: Working on the TWIST Project – AI-based Digital Twin Technology for the diagnosis and prognosis of tidal turbine failures and biofouling. Research focuses on Machine Learning, Deep Learning, Signal Processing, and Computer Vision.	
Master of Science (Artificial Intelligence) FAST NUCES, Islamabad	Jan. 2022 to Jan 2024
• GPA: 3.7/4.0 (Gold Medal) • Coursework: Deep Learning, Image Processing, Machine Learning and Big Data Analytics	
Certified Artificial Intelligence (Professional Training) PIAIC, Islamabad	Aug 2021 to Jan 2023
• Percentile : 87.58 (Grade - A) • Skill Development: Gained practical skills to solve complex, real-world AI problems.	
Bachelor of Science (Software Engineering) SZABIST, Islamabad	Sept. 2017 to July 2021
• GPA: 3.26/4.0 • Coursework: Web Development, Software Engineering and Artificial Intelligence, Research Methodology	
Intermediate of Computer Science PUNJAB COLLEGE, Islamabad	Aug. 2015 to July 2017
• Marks: 749/1100	
Matriculation IMCB G-6/2, Islamabad	Aug. 2013 to July 2015
• Marks: 599/1050	

Experience

FAST NUCES , Lab Instructor & Visiting Lecturer	Isb, Pakistan
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- Conducted Lab such as Artificial Intelligence, Data Mining, Data structures and Fundamental of Computing
- Developing course materials, including lab lectures, lab manuals, assignments, and exams, and employing innovative teaching methods to promote active learning and engagement.
- Facilitate student learning through hands-on activities, lab projects, and class discussions, and provide personalized feedback on student work to support their progress.
- **Researcher - AIM Research Lab** (Jan 2023 to Present)
- Supervised and guided interns working on projects focused on Artificial Intelligence applications
- **Teaching Assistant - Research methodology** (Sept 2023 to Jan 2024)

Sept 2023 to Present
21 Months

Freelancer Fiverr & Upwork

Remote

- Delivering projects in Machine Learning and Data Science.
- Delivered high-quality data annotation and preprocessing services for AI and ML pipelines.
- Designed and implemented interactive dashboards and reports using Tableau for data visualization
- Offered WordPress development services, including website customization
- Executed various graphic design projects, such as logo creation, banners, and UI elements.

Dec. 2019 to Feb. 2024
51 months

SZABIST, Lab Instructor (Visiting)

Isb, Pakistan

- Conducted Data Structures BS Labs covering different sorting and search algorithms. The labs also focused on implementing different data structures in C++.
- Developing course materials, including lab lectures, lab manuals, assignments, and exams, and employing innovative teaching methods to promote active learning and engagement.
- Facilitate student learning through hands-on activities, lab projects, and class discussions, and provide personalized feedback on student work to support their progress.

Dec. 2023 to Jan. 2024
4 months

Internships

Isb, Pakistan

- PHP Laravel Internship at **Macro Soar** (23 Feb 2021 to 23 May 2021)
- Data Science and Business Analytics at **Sparks Foundation** (Feb 2022 to Mar 2022)
- BDO at **Ankasky** (June 2020 to July 2020)

Projects

Activity Recognition

- Developed a Computer vision-based Activity Recognition in COstruction.
- Used LSTM and CNNs

DeepFake Detection(MS thesis)

- Developed a deep fake detection system using lighting inconsistencies in retinal and face movements.
- Implemented landmark Detection.
- used FacedSwapGAN for deep fake detection.

Camera-image based automated stationery calculator using the YOLOv5 algorithm

- Developed an AI-based Automatic Stationary billing system using YOLO.
- Used Computer Vision algorithms along with Python programming language.

Image Steganography

- Implemented the Enhancement of Encoded Cover in Multi-Image Steganography using Deep Learning
- Used deep learning frameworks.

Smart YouTube Video Assistant

- Developed a Smart YouTube Video Assistant that allows users to input a YouTube video URL and a custom prompt, generating relevant responses in various formats (plain text, text with images, video snippets, or a video response).
- Enhanced user experience by providing additional recommendations, including related web sources, images, and YouTube videos.

Intelligent Health Monitoring System (FYP BS)

- Implemented an intelligent Health Monitor System as a final-year project.
- **Framework** : Django python
- **Front End** : HTML, CSS, Bootstrap, JavaScript, High Charts API
- **Back End** : Python, Regression Algo
- **Database** : PostgreSQL
- **Role** : Front End Development, Backend Development, Forecasting Using Artificial Intelligence

Other Projects: Content-Based Image Retrieval, Font Identification, Exam schedule generator using genetic algorithm, Facial emotion recognition, Dementia Classification

Additional Courses and Workshops

- MTA 98-381 Python Certification (Microsoft)
- AI for everyone Coursera Certification (Coursera)
- Workshop attempt on Agile Methodology
- Computer Vision and Image Processing (Deeplearning.ai)
- AI for Medical Diagnosis (Deeplearning.ai)
- Generative AI for Everyone (Deeplearning.ai)
- Introduction to Large Language Models (Google Cloud)
- Bachelor English Proficiency Certificate
- Master English Proficiency Certificate
- French for Beginners

Technologies

Languages: Python, PHP, SQL, C, C++

Front End Development Techniques: HTML 5, CSS, Javascript, Bootstrap

Frameworks: Tensorflow, PyTorch, NLTK, scikit-learn, Django

Tools and Technologies: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Gen-

erative AI, Tableau, Hugging face, OpenCV, mediapipe, Dlib

Other Technologies: Git, Jupyter Notebook, Google Colab, Overleaf

Awards & scholarship

- Award of IGNITE NGIRI grant to FYPs 2020-21
- Gold Medalist MSAI Degree (Issued by National University of Computer & Emerging Sciences)
- Fully Funded PhD Scholarship at Université de Bretagne Occidentale

Dataset Contribution

- AI-Based Automatic Stationery Billing System
- Construction Activity Recognition

Languages

- English (Fluent)
- Urdu (Native)
- French (Basic)

Article & Research Reports

- E. Ali and M. Ishtiaq, "Deepfake detection using lighting inconsistencies in retinal face movements (Under Working)
- "Intelligent Health Monitoring System Using Deep Learning : Ehsaan Ali, Daim Attiq : Free Download, Borrow, and Streaming : Internet Archive," Internet Archive, 2025. Link (accessed Feb. 16, 2025).

Volunteering

- Marketing For Comhec (Marketing for Inter-University Computer Hardware Projects Exhibition) at Shaheed Zulfikar Ali Bhutto Institute of Science and Technology

REFERENCES

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